Social Welfare Under Heterogeneous Time Preferences

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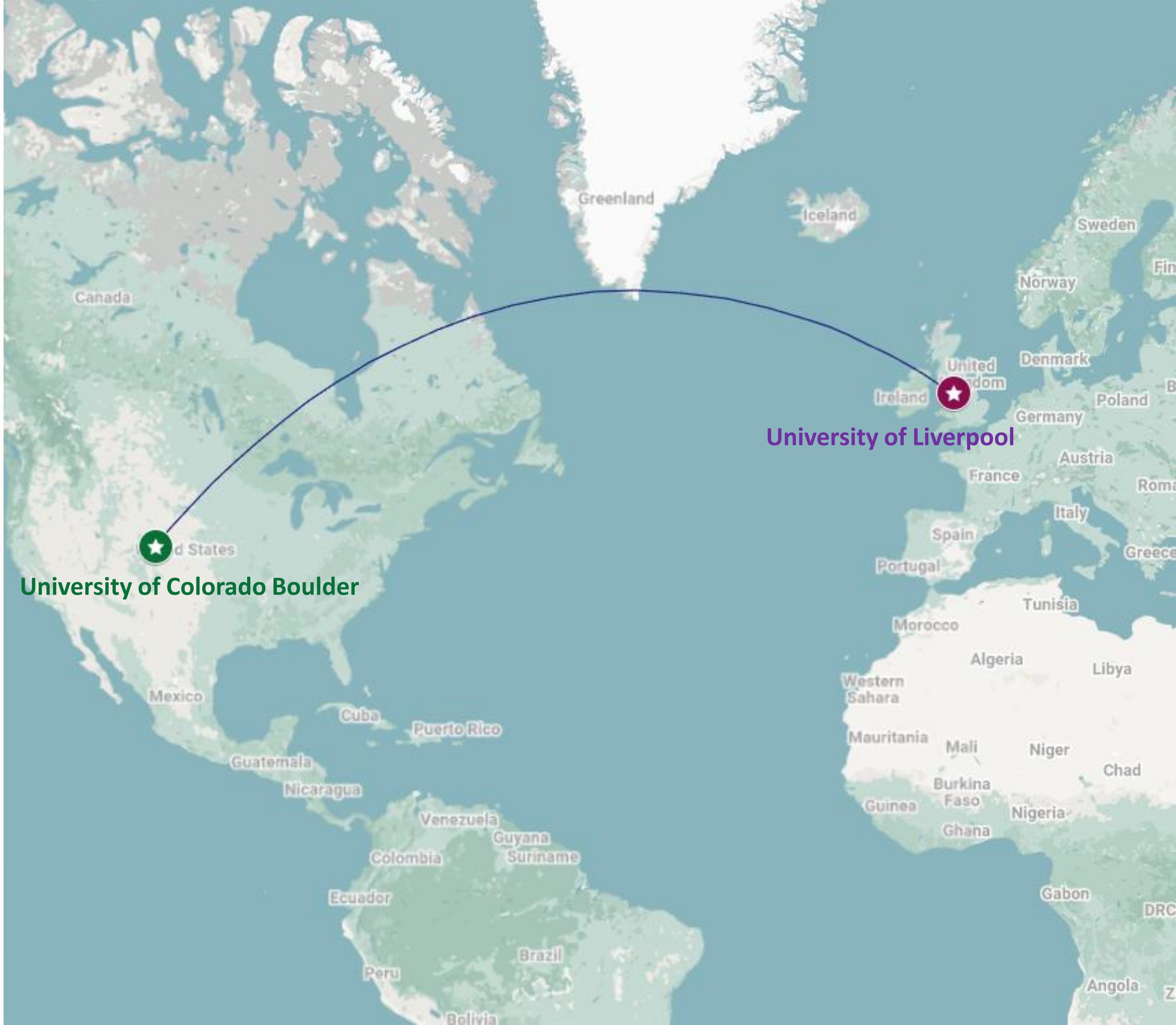


Why Heterogeneous Time Preferences?

Key Question

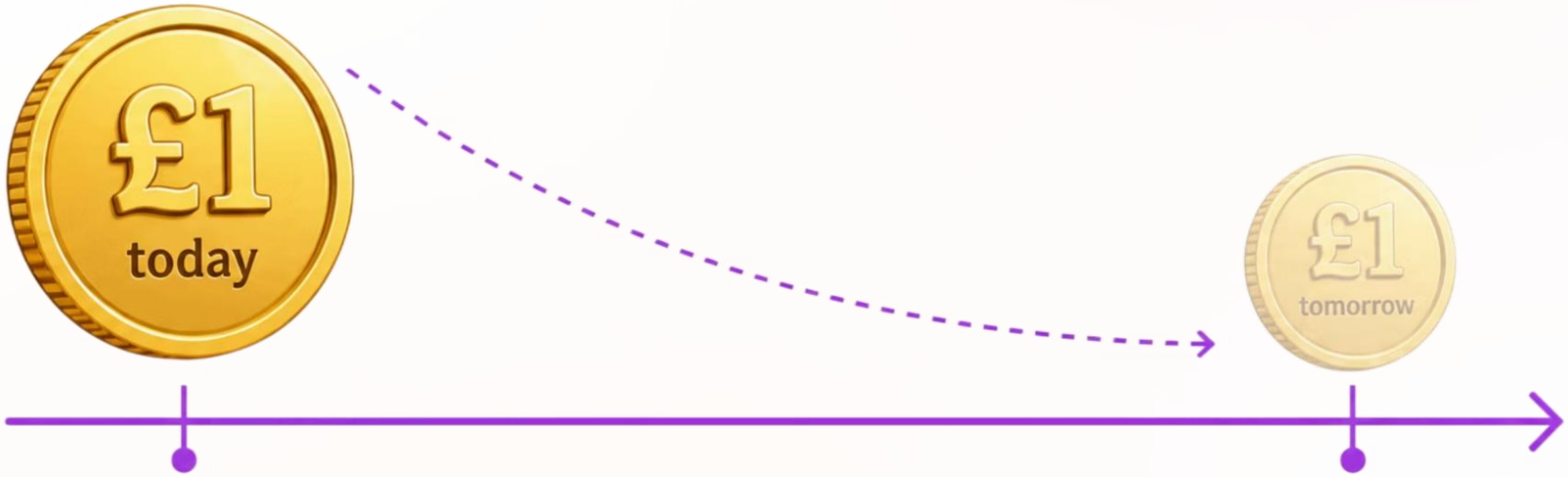
What should an agent do when everyone agrees on the rewards, but disagrees about how much the future is worth?

Climate policy

Infrastructure

Investment

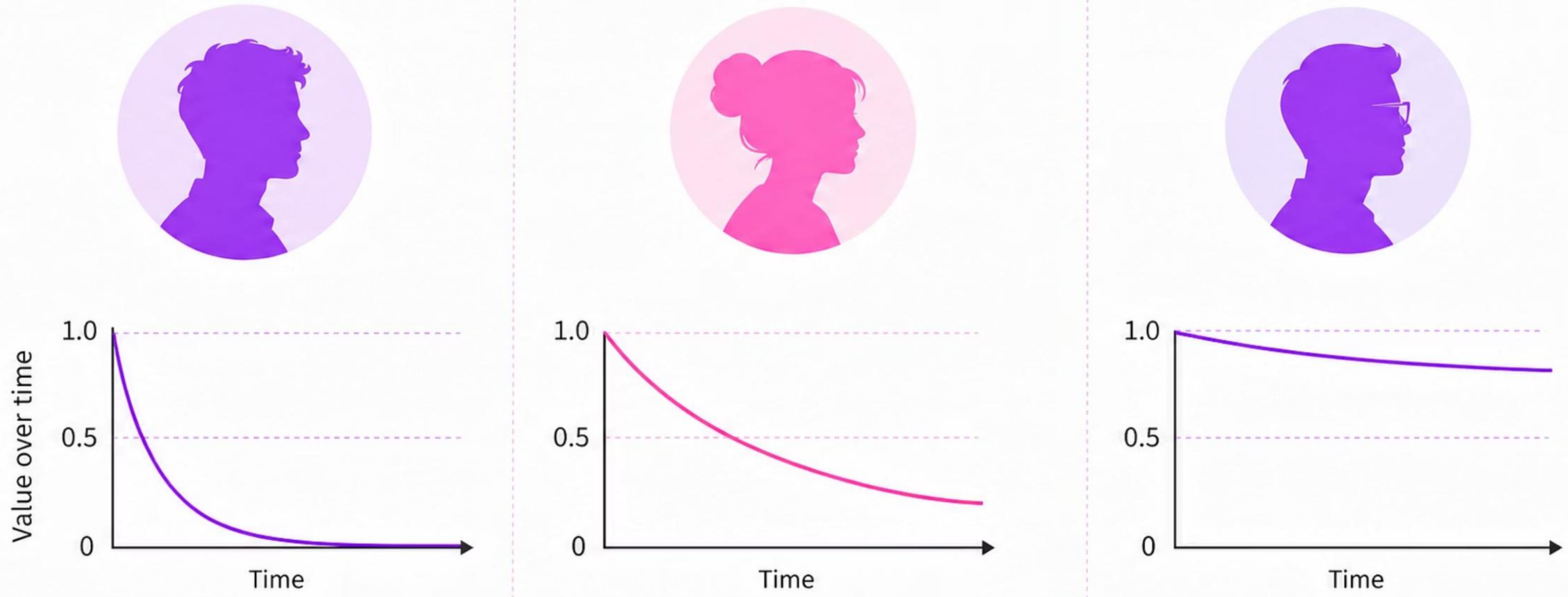
The Value of Delayed Rewards



A reward becomes **less valuable** when it is delayed.

Do People Value the Future Differently?

Yes! People value the future differently.



Low discounting
(impatient)

Moderate discounting

High discounting
(patient)

Classical Setting

Different principals may have different rewards but **same** discount factor.

Our Setting

Principals may also discount future rewards **differently**.

One Decision-Maker

Agent , Decision maker

Multiple Principals

Principals, stakeholders

Social Welfare Objective

Principal i's payoff under policy π

$$V_i^{\pi} = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \lambda_i^t \, R_i^t\right]$$

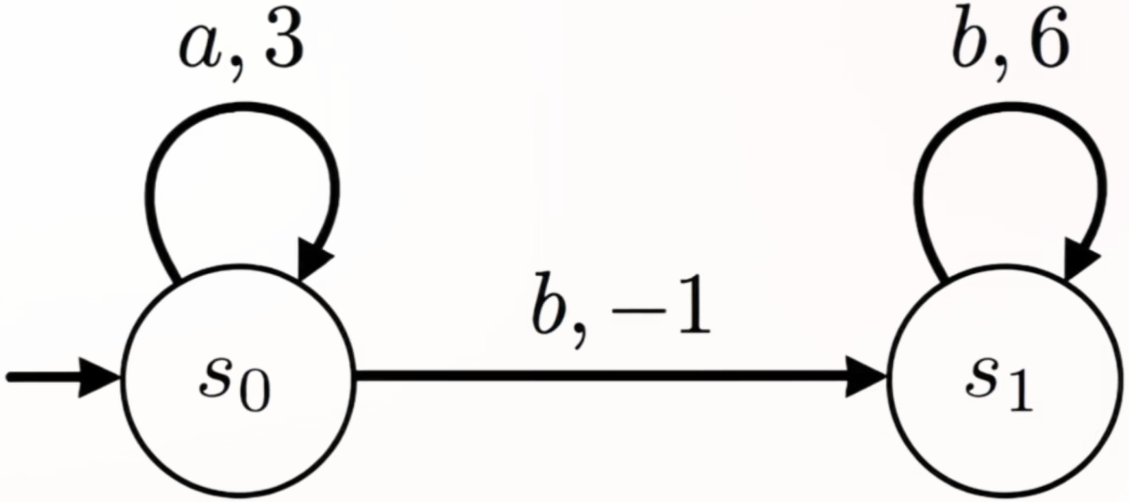
Social Welfare

$$\mathrm{SW}(\pi) = \sum_i V_i^{\pi}$$

Optimisation Objective

$$\max_{\pi} \mathrm{SW}(\pi)$$

Alice and Bob, Open a Restaurant!



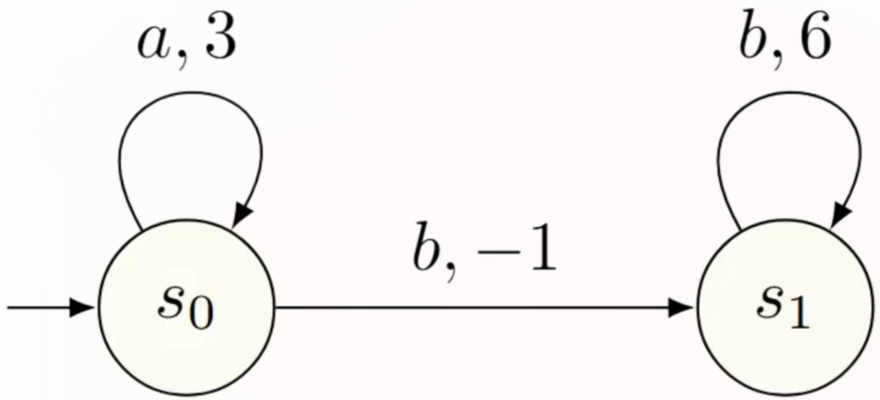
Patient investor

Alice **λ: 2/3**

Impatient investor

Bob **λ : 1/3**

Social Welfare in different strategies



Policy 1: a^ω

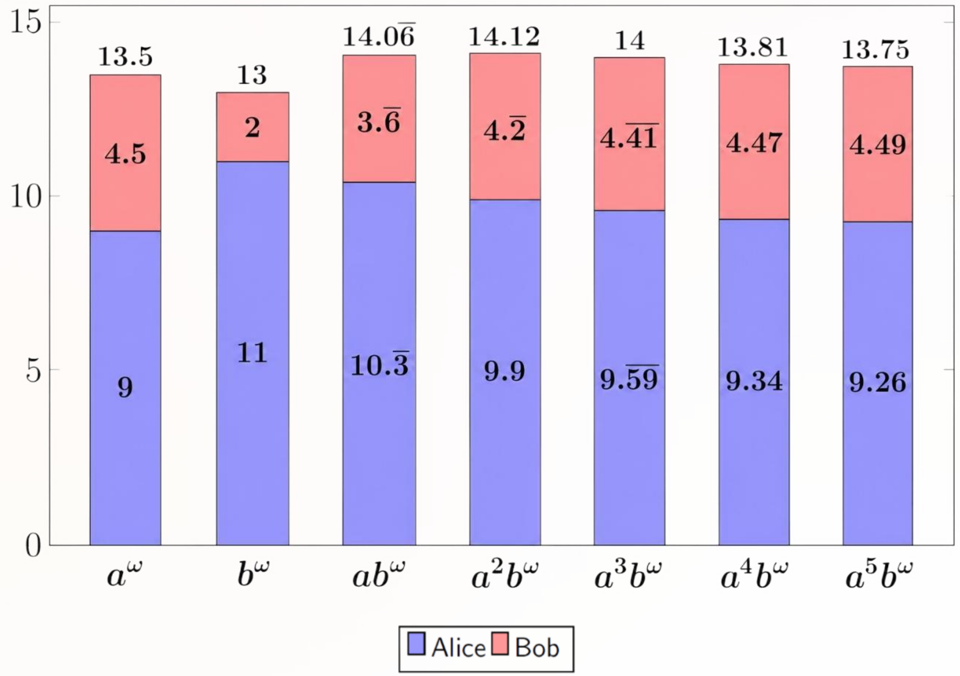
individual payoff: $\dfrac{3}{1-\lambda}$

Policy 2: b^ω

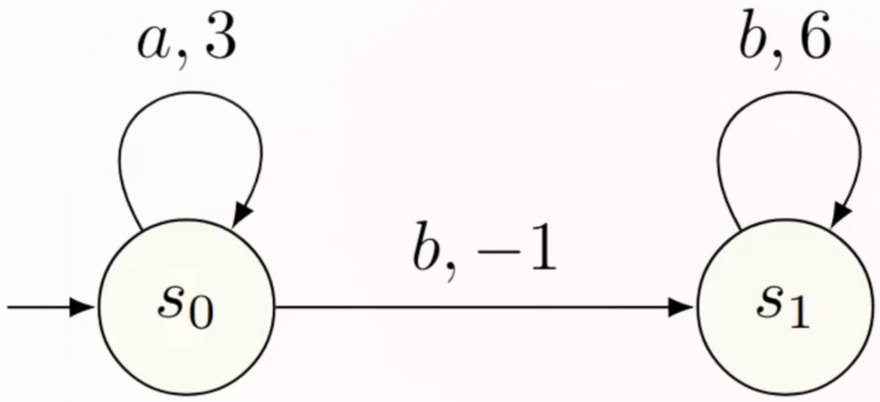
individual payoff: $-1 + 6\dfrac{\lambda}{1-\lambda}$

Policy 3: $a^k\ b^\omega$

individual payoff: $3\dfrac{1-\lambda^k}{1-\lambda} - \lambda^k + 6\dfrac{\lambda^{k+1}}{1-\lambda}$



Social Welfare in different strategies



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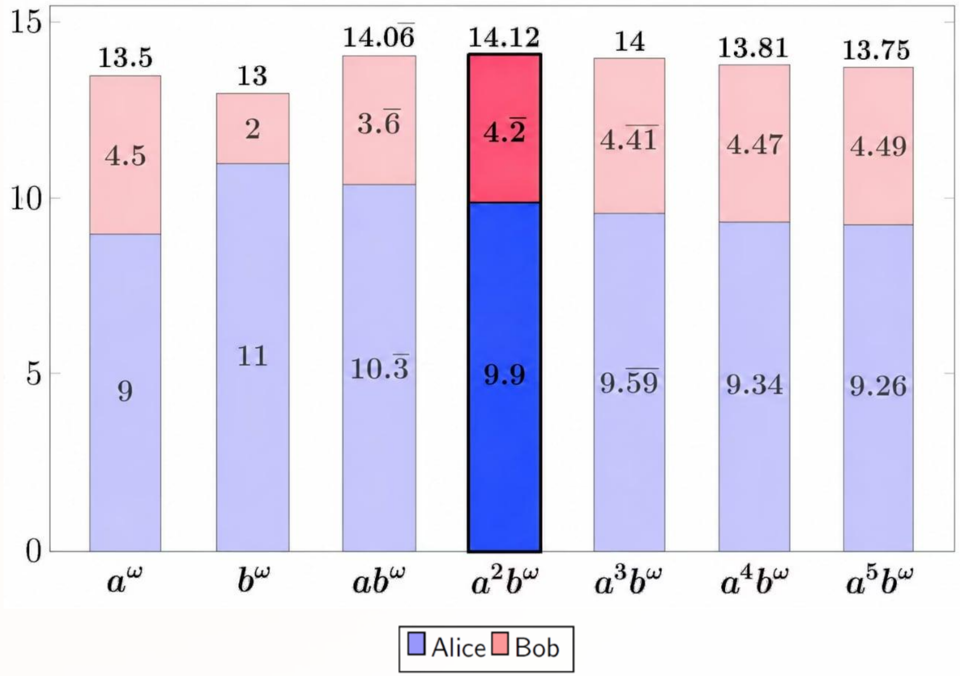
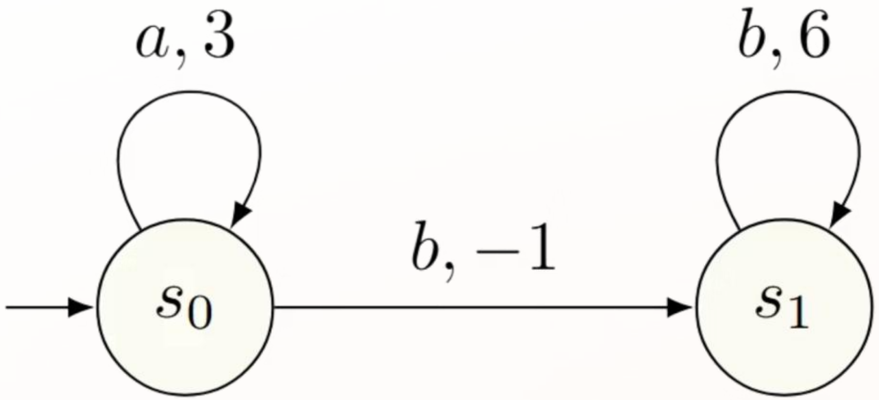
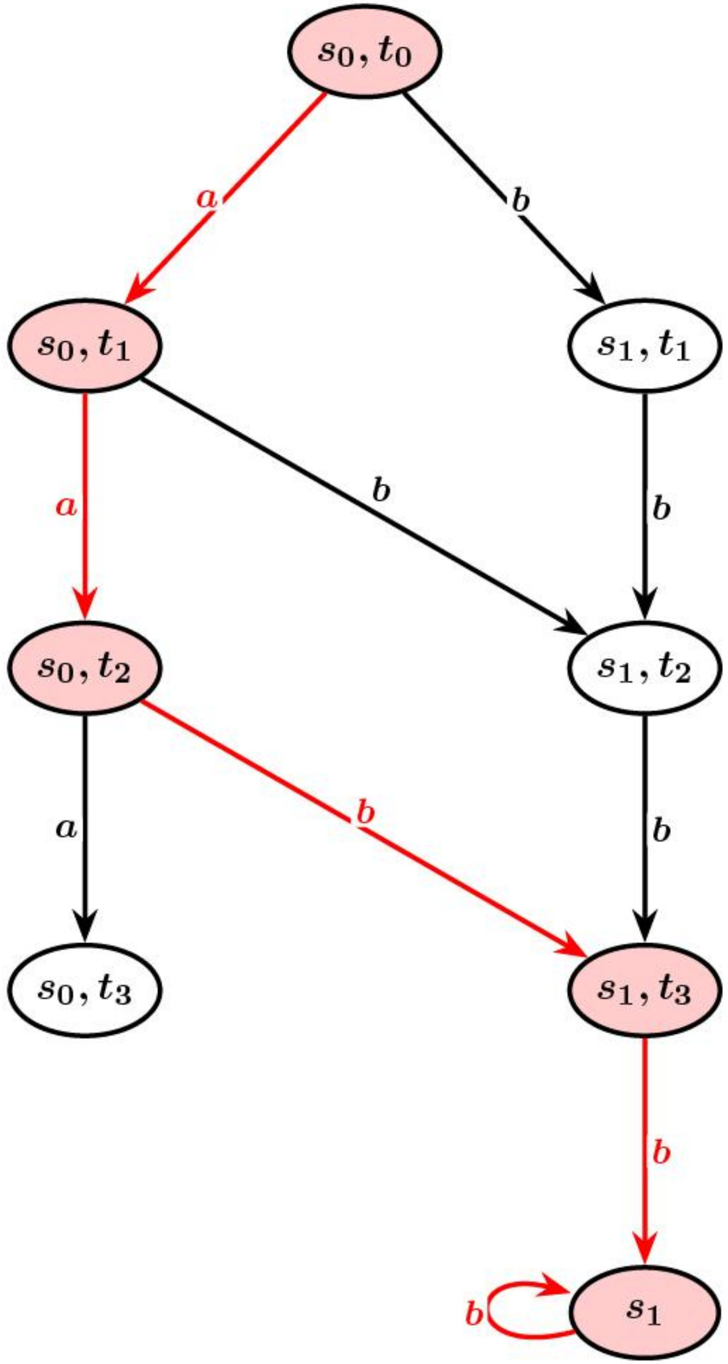
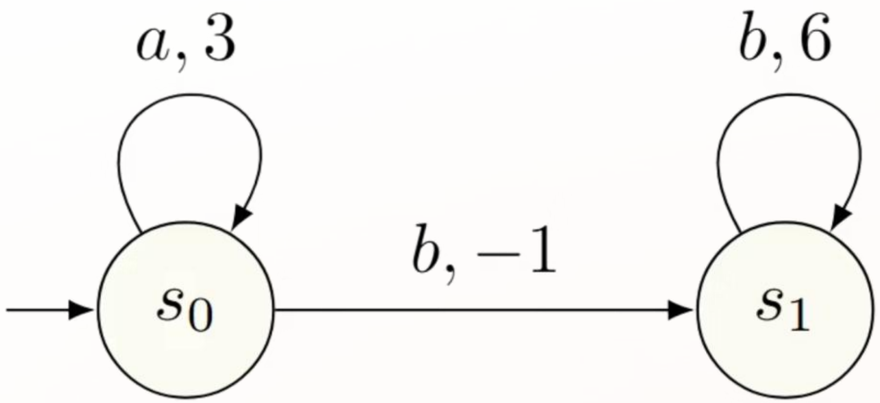


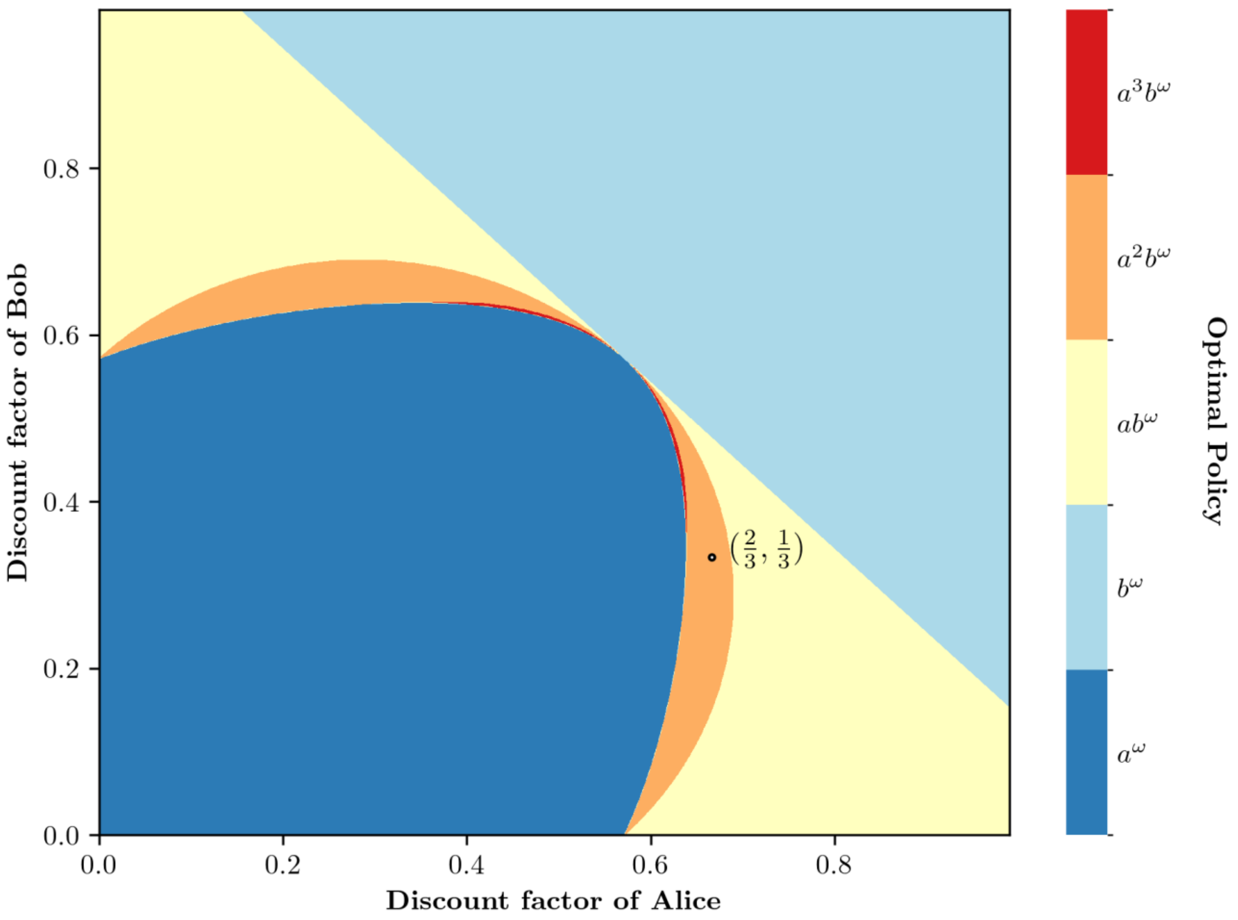
Social Welfare
in different strategies





Heterogeneous discounting turns time itself into **Strategically relevant memory.**





How to get the best Social Welfare?

Negative Results

- Pure stationary strategies?
- Randomised strategies?

Positive Results

✓ Optimal strategies exist in a simple class.
✓ Pure finite-memory counting strategies

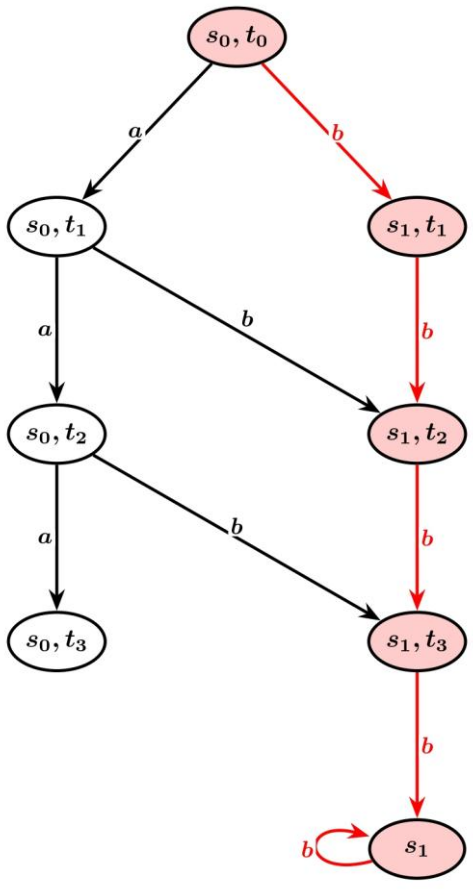
Simple stationary behavior is Hard

Structured finite memory is easy.

Long-run Behavior: Lexicographic Tail

Order principals by patience: $\lambda_0 > \lambda_1 > \lambda_2 > \lambda_3 > \cdots > \lambda_n$

1. **Optimize the MDP for the most patient principal.**
   a. **Restrict to actions that are optimum.**
2. **Break ties using the next most patient principal.**
3. **Continue until last principals.**



M_0 original MDP → M_1 optimal for P_0 → M_2 tie-break P_1 → . . . → M_n long-run tail

Deviation from Long Term Strategy

$\mathrm{M_n}$
long-run tail

For Principal$_\mathrm{i}$:

$$\Delta_0(s,a,i) = R(s,a,i) + \lambda_i \sum_{s'} p(s' \mid s,a) V_i(s') - V_i(s)$$

After *t* steps

$$\Delta_t(s,a,i) = \lambda_i^t \Delta_0(s,a,i)$$

Social effect

$$\sum_i \Delta_t(s,a,i) = \sum_i \lambda_i^t \Delta_0(s,a,i)$$

After a finite horizon, deviations cannot help

M_n
long-run tail

for every deviation, there exists a finite horizon k such that, for all $t \geq k$:

$$\Sigma_i \, \Delta_t(s, a, i) \geq 0$$

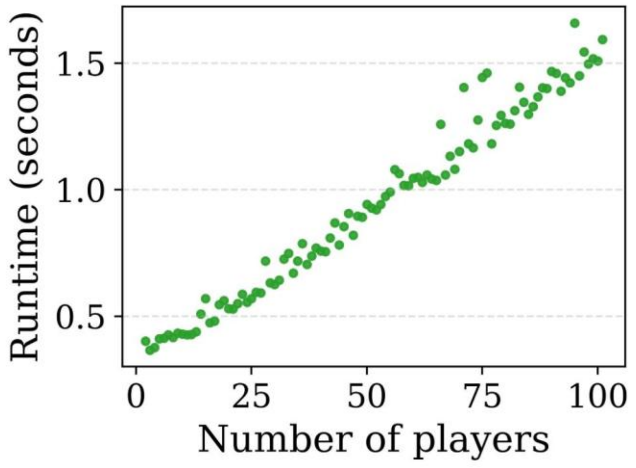
Why κ is small under spacing?

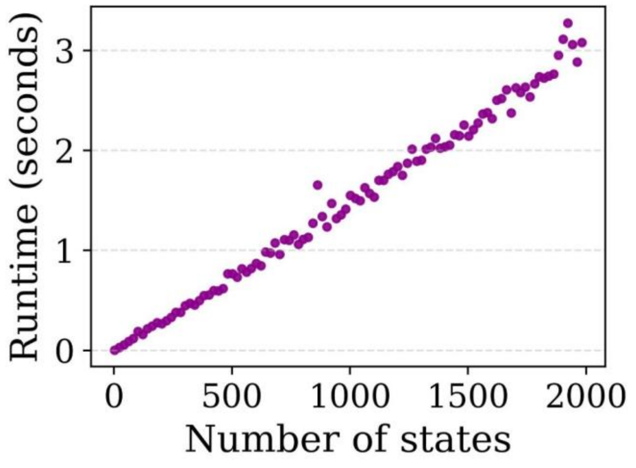
A typical bound has the shape $k = O\left(\frac{\log(\text{value ratio})}{\log(\lambda_i/\lambda_{i+1})}\right)$

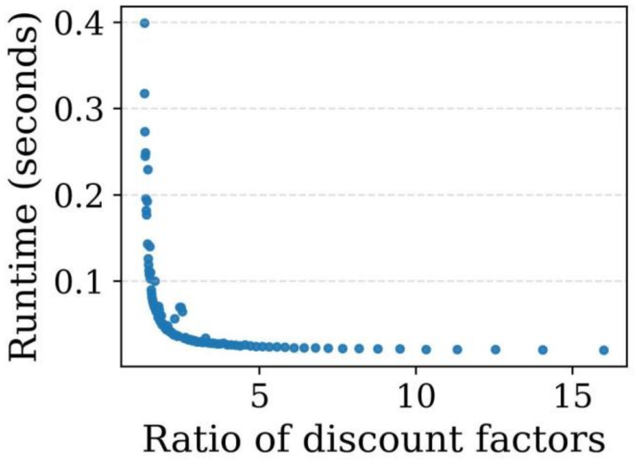
Outcome

Only the prefix before K needs time-dependent optimization; after that, follow the stationary tail.

Scalability Trends







Synthesis Algorithm

Original MDP → Compute lexicographic tail → Find flipping point (k) → Unroll to DAG → Backward induction

Deviation rewards: $$R_{\mathrm{k}}^{\Delta}(s,a) = \sum_{i} \lambda_i^{\mathrm{k}}\, \Delta_0(s,a,i)$$

$$\pi^* = \underbrace{\text{[finite prefix]}}_{\text{counting memory}} \cdot \underbrace{\left[\pi_{\text{long}}^{\omega}\right]}_{\text{Stationarytail}}$$

Complexity Contrast

Insist on stationary simplicity	Allow structured finite memory
▪ Loses social welfare	✓ Attains optimal social welfare
▪ Randomization may help but does not solve the structural issue	✓ Pure finite counting strategies are sufficient
▪ Stationary/positional threshold is NP-hard	✓ Computable in polynomial time

- Simple stationary behaviour is too restrictive

✓ Structured finite memory is enough

Takeaways

Thank you!

1. Heterogeneous time preferences change the temporal structure of optimal control.
2. Socially optimal behavior may need memory even when rewards are identical.
3. Optimal strategies have a simple form: $\pi^* = \underbrace{[\text{finite prefix}]}_{\text{counting memory}} \cdot \underbrace{\left[\pi^{\omega}_{\text{long}}\right]}_{\text{Stationarytail}}$
4. Under reasonable spacing, synthesis is polynomial time.

Different notions of patience do not merely reweight rewards;
they change the structure of optimal behavior.

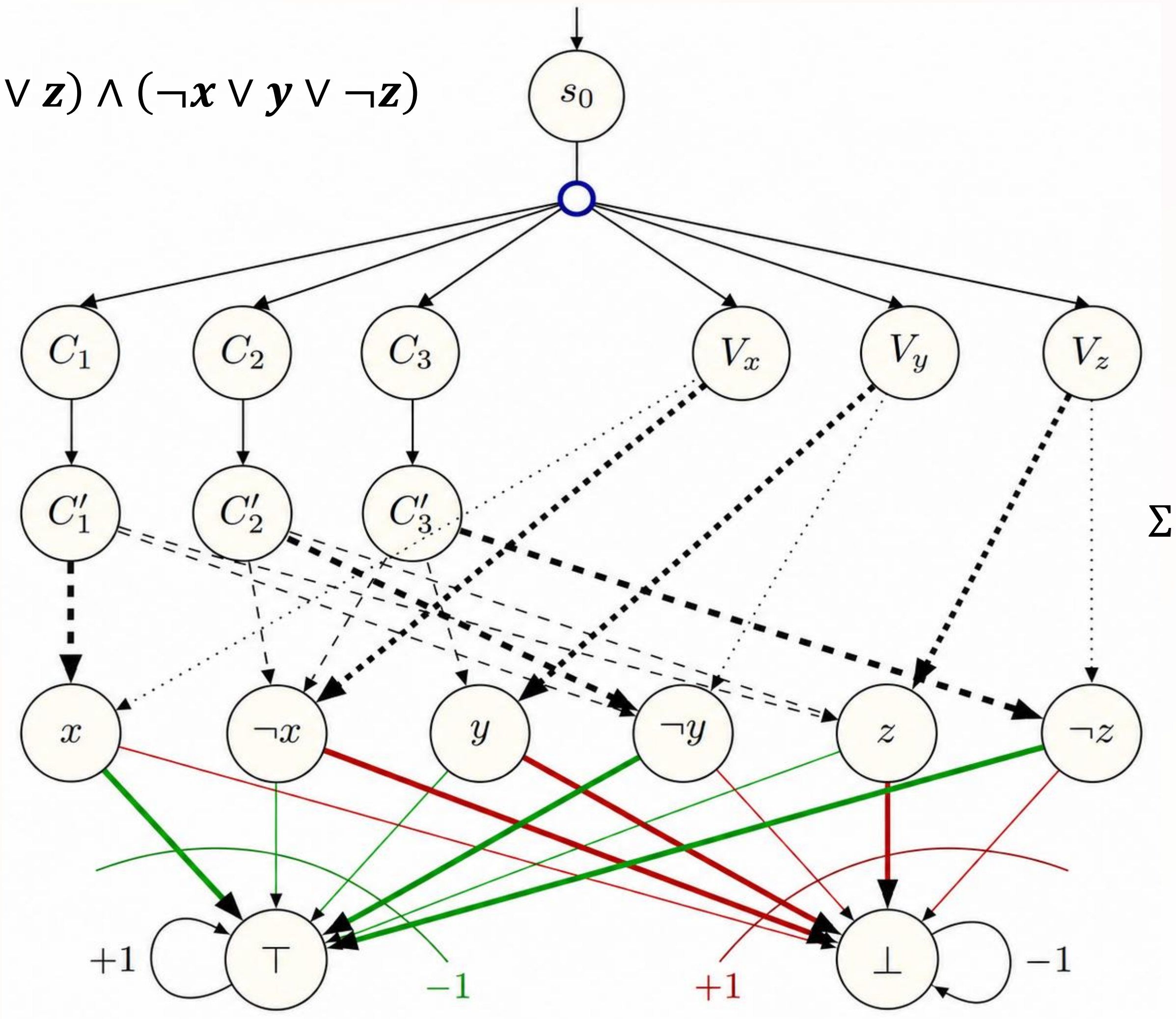
NP-Hardness

$$\boldsymbol{\phi} = (\boldsymbol{x} \vee \neg \boldsymbol{y} \vee \boldsymbol{z}) \wedge (\neg \boldsymbol{x} \vee \neg \boldsymbol{y} \vee \boldsymbol{z}) \wedge (\neg \boldsymbol{x} \vee \boldsymbol{y} \vee \neg \boldsymbol{z})$$

$\lambda_0 = 0.54$

$\lambda_1 = 0.40$

$$\Sigma_i \left(0 + 0 \cdot \lambda_i + 0 \cdot \lambda_i^2 - {\lambda_i}^3 + \frac{\lambda_i^4}{1 - \lambda_i} \right)$$



$$\Sigma_i \left(0 + 0 \cdot \lambda_i + {\lambda_i}^2 - \frac{\lambda_i^3}{1 - \lambda_i} \right)$$